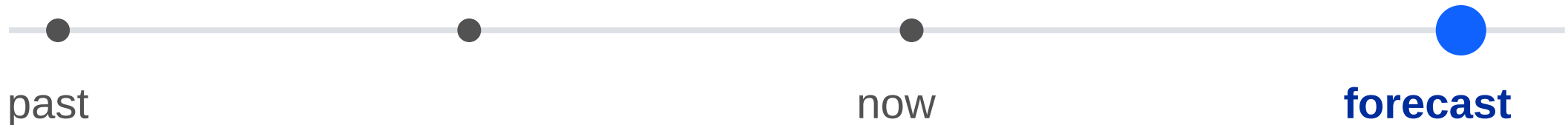


CE436 / PROJECT PROPOSAL

TEMPERATURE FORECAST ON FPGA

+3 h from past hourly values



3 students • 12 weeks • C++ + Verilog/SystemVerilog

A testable forecasting task

INPUT

Hourly temperature
at one location



OUTPUT

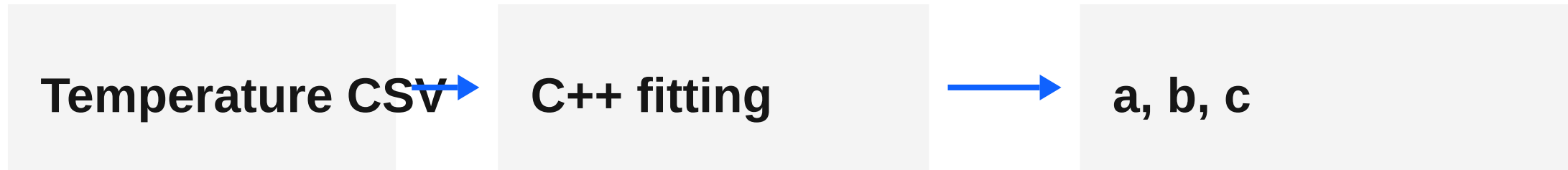
One temperature
3 hours ahead

Known through t

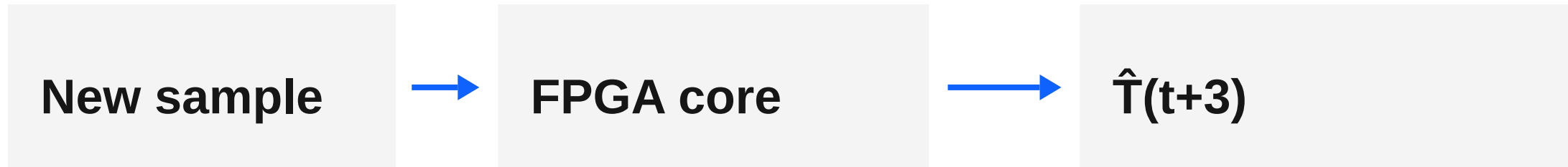
Forecast for $t + 3 \text{ h}$

C++ trains. FPGA forecasts.

OFFLINE ON PC

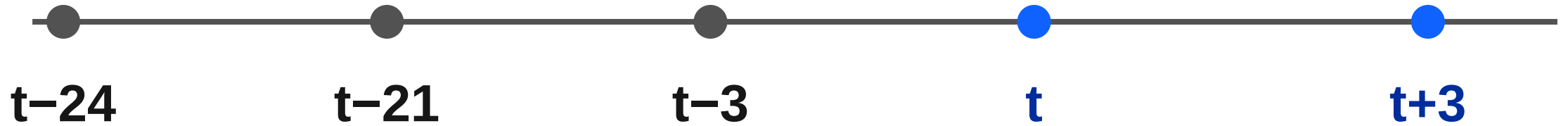


RUNTIME ON FPGA



The coefficients stay fixed while the FPGA runs.

Four past values. One future target.



$$t + 3 - 24 = t - 21$$

25 consecutive samples are stored; the future target is never an input.

A linear model from past values

$$\begin{aligned}\hat{T}(t+3) = & T(t-21) \\ & + a[T(t)-T(t-24)] \\ & + b[T(t)-T(t-3)] & + c\end{aligned}$$

ANCHOR

same hour yesterday

DAY-TO-DAY

current vs. yesterday

RECENT CHANGE

latest 3 hours

How is one forecast computed?

ILLUSTRATIVE VALUES — NOT A MEASURED RESULT

19°	21°	18°	20°
t-24	t-21	t-3	t

$21 + 0.5 \times$
 $\frac{1}{2} \times 0.2 \times$
 2

subtract → multiply → add

→ **21.9°C**

Train on the past. Test on the future.

$$x1 = T(t) - T(t-24)$$

$$x2 = T(t) - T(t-3)$$

$$y = a \cdot x1 + b \cdot x2 + c$$

fit a, b, c by least squares

01 TRAIN

fit coefficients

02 VALIDATE

select settings

03 TEST

evaluate once

Split chronologically; training labels must not cross into later periods.

Does the model beat simple forecasts?

CURRENT VALUE

$T(t)$

SAME HOUR YESTERDAY

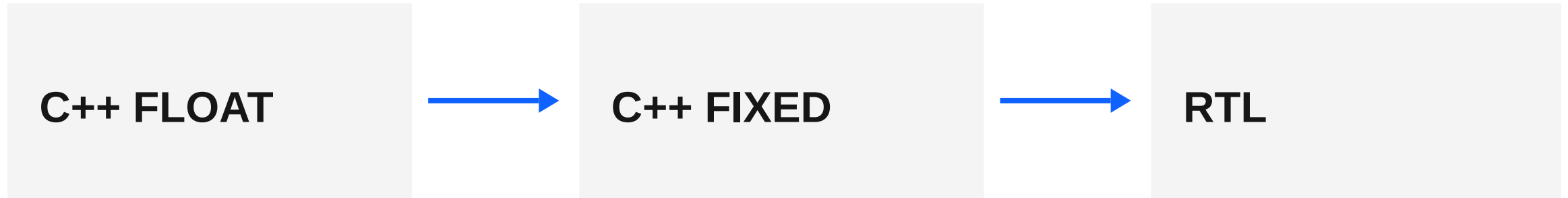
$T(t-21)$

LINEAR MODEL

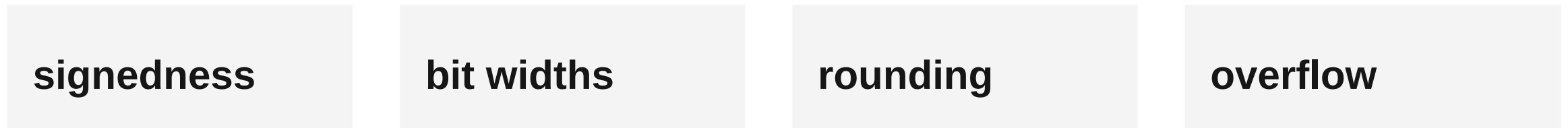
$\hat{T}(t+3)$

Compare MAE on the same held-out test timestamps.

Fixed-point needs an explicit contract

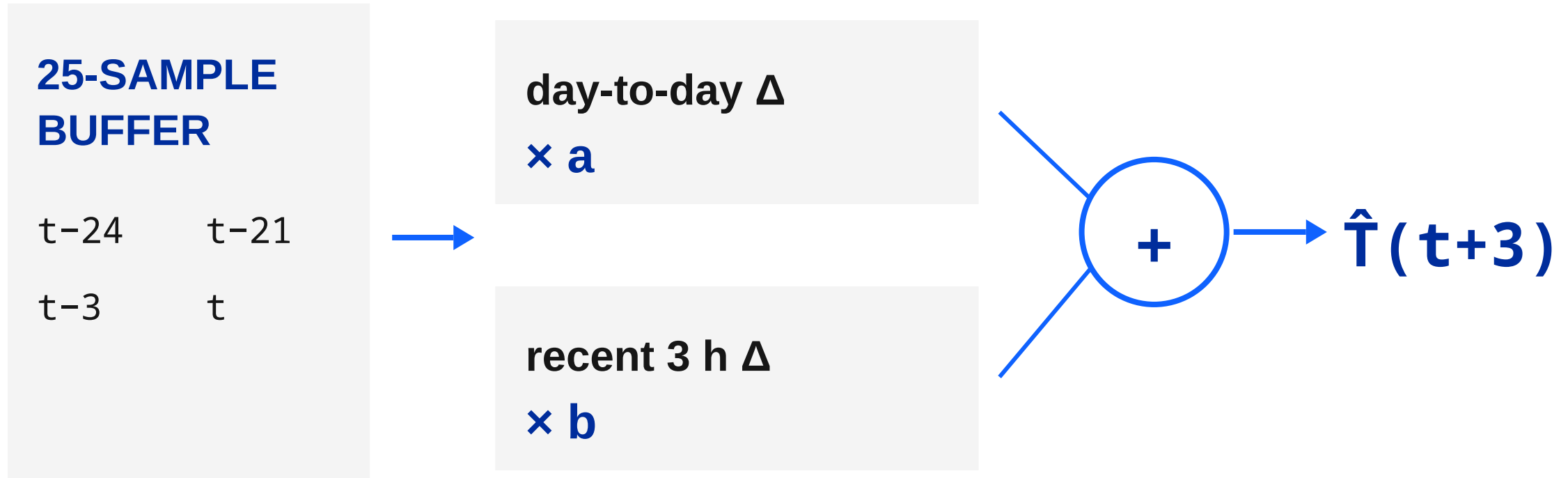


Define the arithmetic before coding RTL:



Forecast error and quantization error are different measurements.

From sample history to arithmetic



Accept a sample on valid; output only after the buffer is ready.

Three verification layers

01

C++

float vs. fixed

quantization

02

TESTBENCH

RTL vs. reference

bit-exact output

03

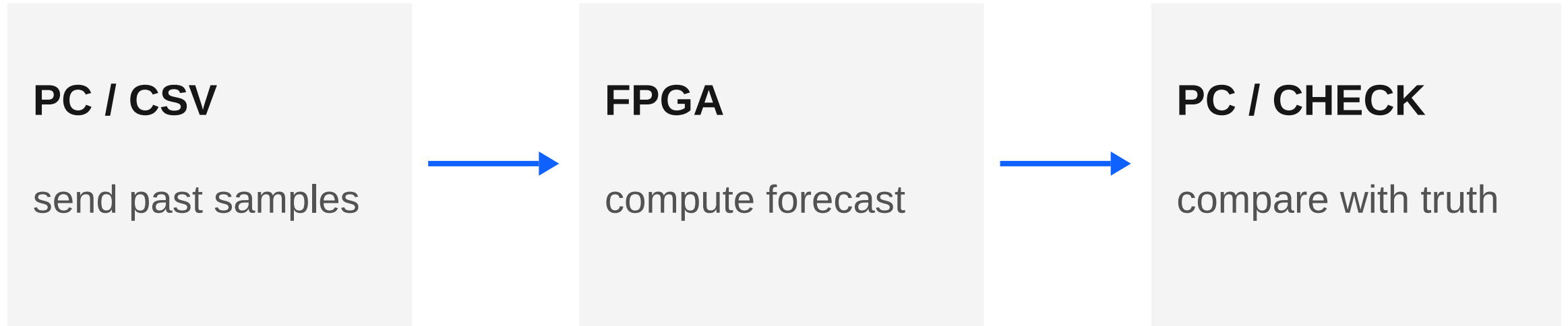
BOARD

integrated run

I/O and sample order

All layers use the same input vectors and arithmetic specification.

Replay historical data for the demo



Only samples through t enter the FPGA.

12 weeks. Prototype early. Evaluate deeply.

W1–3

PROTOTYPE

data + C++
working RTL core

W4–6

VALIDATE

model vs. baselines
fixed-point sweep

W7–9

HARDEN

bit-exact tests
resource trade-offs

W10–12

DEMO

board integration
final test + report

Each phase ends with a measurable deliverable.

Three students. Clear ownership.

STUDENT 1

DATA + C++

dataset + MAE
coefficients

missing hours

STUDENT 2

FIXED + RTL

RTL core
synthesis

board TBD

STUDENT 3

VERIFY + DEMO

test vectors
interface + logs

high error

One shared contract: data, coefficients, rounding, valid, and latency.

One forecast. Two things to prove.

01 FORECAST QUALITY

Beat the baselines
on unseen data

02 HARDWARE CORRECTNESS

Match the C++ fixed-
point reference

Feedback needed:

scope • FPGA board/tools • minimum demo